

technology are creating and testing various applications for it, such as the following.

- **Finance and transaction processing on the blockchain:** Blockchain based transactions could be resolved in seconds, and bank transfer costs drastically reduced or eliminated.
- **Supply chain tracking with blockchain technology:** Blockchain technology may help companies rapidly identify areas for improvement in their supply chains, track inventory in real-time and monitor product quality from the factory to the store shelf.
- **Blockchain for electronic credentials:** Microsoft is exploring the use of blockchain technology to give consumers more control over their digital identities and the information associated with them.
- **Using blockchain to distribute information:** Blockchain has the potential to serve as an intermediary for the safekeeping and transfer of business data between sectors.
- **Protection of intellectual property and royalties using blockchain:** By using blockchain technology to construct a distributed ledger, we can guarantee that musicians will always own their work and distribute royalties in an open and timely manner. Similar benefits may accrue to the open source community if it uses blockchain technology.
- **Healthcare blockchain:** Blockchain technology may potentially have significant applications in the healthcare sector. It is being used by healthcare providers to keep track of data received from clinical trials and electronic medical records.

So, how does the Blockchain function?

There are three key components to a blockchain: blocks, nodes, and miners.

Why we use blocks: Multiple blocks make up every chain, and all have the same three essential components:

- In the block, data.
- A 'number used just once', or 'nonce'. In blockchain, a nonce is a random 32-bit whole integer that is used to build a block header hash.
- In blockchain technology, the 256-bit hash that is integrally tied to the nonce must start with a high number of zeros (i.e., be extremely small). A nonce is used to produce the cryptographic hash of the first block in a chain. Until the block is mined, the data inside it is signed and inextricably linked to the nonce and hash.

In the world of blockchain, what exactly is a miner? 'Mining' refers to the method through which new blocks are added to the chain.

Mining a block in a blockchain is difficult, since each block has its nonce and hash, and contains a reference to the hash of the prior block in the chain. This is particularly true for big blockchains.

Miners utilise specialised software to tackle an extremely difficult mathematical problem to determine a nonce that results in a valid hash. A 256-bit hash must be mined with a 32-bit nonce, with about four billion potential nonce-hash combinations. When this occurs, miners have identified the 'golden nonce', and their block is added to the blockchain.

When a block is mined successfully, all nodes in the network agree to the new data, and the miner receives payment.

What does decentralisation in blockchain mean? Blockchain technology relies heavily on the idea of decentralisation. Nodes in a blockchain network may be any kind of computer or other electronic device that stores a copy of the blockchain and facilitates the network's continued operation. As a result, the chain cannot be controlled by any central computer or authority. Instead, it is a decentralised database accessible from any network node.

Every transaction in a blockchain may be readily verified and observed by anybody. Each node stores its own copy of the blockchain, and for the chain to be updated, trusted, and confirmed, the network must algorithmically approve each new block that is mined. Each user is assigned a unique number that may be used to track their financial dealings inside the system.

The blockchain's integrity is kept intact and trust is built among its users thanks to the combination of public information and a system of checks and balances. Blockchains may be viewed as the scalability of trust via technology.

How safe is blockchain technology?

Multiple features of blockchain technology contribute to the system's decentralised nature and high level of security and trust. The order in which newly added blocks are kept is always chronological. They are always appended to the 'last' part of the blockchain. It is exceedingly difficult to change the contents of a block after it has been appended to the end of the blockchain unless there is widespread agreement to do so. This is because each block stores not only its hash but also the hash of the prior block and the date that was discussed before. Data is converted into a string of alphanumeric characters using a mathematical formula called a hash function. When the data is altered, the corresponding hash code also shifts.

Let's take an example. A hacker who also controls a node on the blockchain network plans to change the ledger to steal Bitcoins from everyone. If each one of them changed their copy, the results would be inconsistent. All other copies would be compared to this one, and the hacker's would be immediately disregarded as invalid since it would stand out as different.

For such a hack to be successful, the attacker would need to take over 51% or more of the blockchain